Worlds, strict conditionals, and variably strict conditionals

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1. Background on propositions and possible worlds

Three important operations on propositions: conjunction (∧), disjunction (∨), negation (¬).

According to a view I call *Booleanism*, propositions form a *Boolean algebra* with respect to these operations. What this means: for any propositions p, q, r

(Commutativity)	(a) $p \wedge q = q \wedge p$	(b) $p \vee q = q \vee p$
(Distributivity)	(a) $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$	(b) $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$
(Absorption)	(a) $p \wedge (q \vee \neg q) = p$	(b) $p \vee (q \wedge \neg q) = p$

It turns out to follow from these axioms that whenever the material conditional '$\phi \equiv \psi$' is a theorem of propositional logic, the proposition that ϕ = the proposition that ψ.

Some further definitions that make sense, given Booleanism:

p **entails** q, or for short '$p \leq q$', iff $p \wedge q = p$, or equivalently iff $p \vee q = q$. (Note that the above axioms entail that if $p \leq q$ and $q \leq p$ then $p = q$.)

p is the **greatest lower bound** of set S: p entails every member of S and is entailed by every other proposition that entails every member of S. (Note that $p \wedge q$ is the glb of {p,q}. The glb of S is also called the *conjunction* of the members of S.)

p is the **least upper bound** of set S: p is entailed by every member of S and entails every other proposition that is entailed by every member of S. (Note that $p \vee q$ is the lub of {p,q}. The lub of S is also called the *disjunction* of the members of S.)

$\bot$ is $p \wedge \neg p$ (for any p — note that the above axioms entail that $p \wedge \neg p = q \wedge \neg q$)

$\top$ is $p \vee \neg p$ (for any p — note that the above axioms entail that $p \vee \neg p = q \vee \neg q$)

A proposition a is an *atom* iff $a \neq \bot$ and for every b, if $b \leq a$, then $b = a$ or $b = \bot$.

A further hypothesis (that makes life very easy): the propositions form a *complete atomic* Boolean algebra. What this means:

(Completeness) Any set S of propositions has a greatest lower bound and a least upper bound.

(Atomicity) For every $p \neq \bot$, there is an atom a such that $a \leq p$.

Given these further hypotheses, there is a one-to-one correspondence between propositions and *sets of atoms*: each proposition corresponds to the set of atoms that entail it; each set of atoms corresponds to its least upper bound (disjunction). We can identify *worlds* with atoms.

Booleanism is a controversial view. However, it is much less controversial to hold that there is some relation of 'mutual entailment' on propositions, such that the Boolean axioms become true when '=' is interpreted to mean logical equivalence rather than identity. On this view, the set of all propositions does not form a Boolean algebra, but the set of *equivalence classes of propositions under mutual entailment* does form a Boolean algebra. And it still looks pretty plausible that *this* algebra is complete and atomic. If we think it is, we can identify "possible worlds" with the atoms of this algebra. Every proposition will correspond to a set of possible worlds—the set of atoms whose members entail it—and in cases where we don't care about the differences between mutually entailing propositions, we won't go wrong if we just equate the propositions with the corresponding sets of worlds.

- To make this more precise: say that a function f from propositions is *non-hyperintensional* iff, whenever p and q are mutually entailing, f(p) and f(q) are mutually entailing.

For many operations on propositions, it will be plausible to hold that applying the operation to logically equivalent inputs yields logically equivalent outputs. Those kinds of operations will be associated with functions from sets of worlds to sets of worlds; we get the set of worlds corresponding to the result of the operation by applying the function to the set of worlds corresponding to the input. If we only care about identifying the outputs of the operation up to mutual entailment, we will lose nothing by just looking at the corresponding function on sets of worlds.

2. Background on modal logic.

'Necessary' and 'possible' can mean many different things. Think of their meanings as *properties of propositions*, or the associated *functions from propositions to propositions*. We take it that these are non-hyperintensional.

- For each meaning of 'necessary' there is a corresponding meaning of 'possible', such that 'It is necessary that ϕ' is logically equivalent to 'it is not possible that not-ϕ', and 'It is possible that ϕ' is logically equivalent to 'It is not necessary that not-ϕ'. ('Duality').
- 'Necessarily (ϕ_1 and ϕ_2 and … ϕ_n)' is logically equivalent to 'Necessarily ϕ_1, and necessarily ϕ_2, and … and necessarily ϕ_n'. Similarly, 'Possibly (ϕ_1 or ϕ_2 or … ϕ_n)' is logically equivalent to 'Possibly ϕ_1, or possibly ϕ_2, or … or possibly ϕ_n'.
- This is also true for *infinite* conjunctions and disjunctions.

- So, the actions of the functions corresponding to by 'possibly' and 'necessarily' on *arbitrary* sets of worlds is completely pinned down once we are given the action of these functions on *singleton* sets of worlds. The result of applying the 'possibly' function to a set of worlds is just the union (disjunction) of the results of applying the operation to its singleton subsets.
- When the result of applying the possibility-function to {v} contains w, say that v is *accessible from* w, or '*w* sees *v*', sometimes written *w*<*v*. So we have the following:
  - '$\Diamond\phi$' is true at w iff ϕ is true at some world accessible from w
  - '$\Box\phi$' is true at w iff ϕ is true at every world accessible from w
- So: up to mutual entailment at least, a pair of corresponding interpretations of 'necessary' and 'possible' is fully specified by its corresponding accessibility relation.
- We can then go on to further categories modalities by looking at properties of their corresponding accessibility relations, such as reflexivity, symmetry, and transitivity. For example, reflexivity (every world is accessible from itself) corresponds to the claim that 'Necessarily ϕ' entails ϕ and ϕ entails 'Possibly ϕ'.

3. The strict conditional analysis of conditionals

Schematic proposal: the proposition expressed by 'If ϕ, ψ' is equivalent to the one expressed by 'Necessarily, either not-ϕ or ψ', *on some interpretation of "necessarily" or other*.

- In terms of accessibility: there is an accessibility relation such that 'If ϕ, ψ' is true at w iff ψ is true at every ϕ-world that is accessible from w.

The strict conditional analysis lets us avoid the objections to the material conditional analysis considered last week. The 'paradoxes of material implication' are no longer valid; and it can be rational to be much less confident in the truth of 'if ϕ, ψ' than in that of '$\phi \supset \psi$', for the same reason that it can be rational to be much less confident in the truth of 'necessarily ϕ' than in the truth of ϕ.

- Given that we are being so liberal about what counts as a possible interpretation of 'necessarily', the material conditional analysis is a *special case of* the strict conditional analysis—consider the "trivial modality" whose associated accessibility relation is the relation each world bears only to itself. This makes 'necessarily ϕ' equivalent to ϕ.
- If the strict conditional analysis is correct, there is plausibly quite a lot of vagueness and context-sensitivity associated with conditionals: different uses of 'if' will often correspond to different accessibility relations.
- Here are some suggestions for how it might go:

- for a typical use of indicative conditional, v is accessible from w iff v is compatible with what the speaker knows at w. (More generally, we may want to look not just at the speaker but at other contextually relevant people, and we may want to consider times other than the time of speech).
- for a typical use of a counterfactual conditional, v is accessible from w if v is compatible with (i) the laws of nature of w, and (ii) the history of w up to some contextually relevant time.

- From the point of view of a strict-conditionalist, it's plausible to think that certain uses of ordinary modals can pick up on the same accessibility relations that matter for conditionals. For indicatives, the relevant modals would be *epistemic* modals like 'Must' and 'might'/'may'; for counterfactuals, they would be so-called *circumstantial* modals, like 'had to' and 'could have'.

4. Lewis and Stalnaker's objections to the strict conditional account

The following three controversial argument-forms are all valid on the strict condiitonal approach:

> **Antecedent Strengthening:** If A, C; therefore if A and B, C.
>
> **Transitivity:** If A, B; if B, C; therefore if A, C.
>
> **Contraposition:** If A, B; therefore if not B, not A.

Influentially, Lewis and Stalnaker offered counterexamples to these inference rules—at least for counterfactuals—and used them to argue for an alternative to the strict conditional anlaysis.

Counterexamples to antecedent strengthening for counterfactuals:

> If you walked on the grass it would be fine. ? Therefore, if you and everyone else in the building walked in on the grass, it would be fine.

Counterexample to transitivity for counterfactuals (Stalnaker)

> (Stalnaker) If Hoover had been Russian, he would have been a communist. If Hoover had been a communist he would have been a traitor. ? Therefore, if Hoover had been Russian, he would have been a traitor.

Counterexample to contraposition for counterfactuals:

> If I had talked to Mary, I wouldn't have told her anything important. ? Therefore, if I had told Mary something important, I wouldn't have talked to her.